Connector-Native Apps (Remote MCP Server)

Mode 2 Phase 1 (Building Blocks), Chapter 2/6 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/connector-native-apps

EN: 'You own the server, not the mind that calls it.' Build a remote MCP server whose customer is an AI, not a human -- a real product ('Reading Room') with memory, real sign-in, and a session contract, running from one pasted URL in claude.ai.
Roman Urdu: 'Aap server own karte ho, us dimagh ko nahi jo usay call karta hai.' Ek remote MCP server banao jiska customer AI hai, insaan nahi -- real product ('Reading Room') memory, real sign-in, session contract ke sath, claude.ai mein ek pasted URL se chalta hua.

1. Four Non-Negotiables

- **1. One gateway** -- AI gets one connector (free accounts allow only 1). **2. Tools only** -- no resources/prompts, since the model calls them mid-reasoning.
- **3. Prove, don't trust** -- identity always from a verified sign-in, never from what the AI claims. **4. Fail closed** -- if the server breaks, the AI must say 'unavailable', never improvise.

2. State + Domain + Proving Identity

- State lives in Postgres: users (verified sub, email) + user_state (per-user JSON). sub = the verified sign-in ID -- the one piece of identity you can trust.
- **OAuth, 5-line summary:** user signs in elsewhere -> gets a signed token -> server verifies it -> reads sub -> model NEVER supplies identity itself.
- **4 checks** (border desk passport): genuine? trusted issuer? stamped for US (most dangerous to skip)? still in date?

3. Steering the Model + Fail Closed

- Rules live in the connector (Option B, recommended) reinforced 4 ways: tool description, session-init return, each tool's 1-line reminder, token-gated real tools.
- **begin_session** is called first -- returns rules+state+a short-lived session 'keycard'. Every real tool: 'no keycard, no entry.'
- **Fail closed** is the most important rule: if the connector is missing/broken, the model will otherwise silently improvise or invent state. Test: kill Postgres, confirm the app cleanly refuses.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
MCP server	A server following a standard protocol so any AI can use it without custom wiring.	Server jo standard protocol follow karta, taake koi bhi AI custom wiring ke bina use kar sake.
Fail closed	On any break, say 'unavailable' -- never let the AI improvise or invent.	Kisi bhi break par 'unavailable' kaho -- AI ko kabhi improvise/invent na karne do.
sub (subject)	The verified sign-in identity -- the one thing the model can never fake.	Verified sign-in identity -- woh ek cheez jo model kabhi fake nahi kar sakta.
begin_session	The first-called tool returning rules, state, and a short-lived session token.	Pehla-called tool jo rules, state, aur short-lived session token wapas deta.
Stateless streamable HTTP	A transport with no memory of one connection, so any server instance can handle any call.	Transport jisme ek connection ki memory nahi, koi bhi server instance kisi bhi call handle kar sake.

Quick Self-Check

Question	Answer
Why does this architecture only expose 'tools', never resources or prompts?	The model calls tools mid-reasoning on its own; resources/prompts require a human picking something, which breaks the agent-as-customer shape.
Why must the model never supply its own identity?	It can be tricked or lie -- identity must always come from a verified sign-in token's sub field, checked server-side.
What's the single most dangerous OAuth check to skip?	Checking the token was stamped FOR US (audience) -- skipping it allows a token from another app to be replayed here.
What's the test that proves 'fail closed' actually works?	Kill the database/connector and confirm the app clearly refuses instead of inventing a plausible-looking answer.

Where This Leads / Sources

Leads to: Build AI Agents (owning the loop), Plugins for AI Agents (mirror image), AI Identity (own your sign-in). Full 14-concept capstone + ceiling table in 05.

Source: docs/connector-native-apps/ (this repo), content originally via Zia Tutor AI connector.